ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID-SEMESTER EXAMINATION **WINTER SEMESTER: 2021-2022**
DURATION: 1 Hour 30 Minutes **FULL MARKS: 75**

CSE 4303: Data Structures

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 3 (three)** questions. Marks of each question and corresponding CO and PO are written in the right margin.

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1. a) Analyze the following block of code and find out the time complexity showing detailed steps (try to make the upper bound tighter). — 7 CO1 PO1

```
void complexity () {
  int n, val;
  for (int j=4; j<n; j=j+2) {
    val = 0;
    for (int i=0; i<j; ++i) {
      val = val + i×j;
      for (int k=0; k<n; ++k) {
        val++;
      }
    }
  }
}
```

b) Given the following XHTML fragment, show the state of the stack (recording the opening tags that have not yet been matched) at the end of parsing. — 8 CO1 PO1

```
<xhtml>
<body>
<p>1b. if lg(<i>n</i>) = log<sub>2</sub>(<i>n</i>) = <i>x</i> then what
is log<sub>16</sub>(<i>n</i>) in terms of <b>x</b>?</p>
<p>1c. Show that lg(2<i>n</i>) = 1 + lg(<i>n</i>) and that ln(2<i>n</i>)
= ln(2) + ln(<i>n
```

c) Show the step-by-step process to sort the following set of number in descending order using Heap sort algorithm: $\{22, 55, 36, 44, 100, 45, 12, 63, 1, 89\}$ — 10 CO1 PO1

2. a) Showing necessary mathematical arguments prove that, 'A max-heap can be formed from an arbitrary set of numbers in linear time'. — 8 CO3 PO2

b) Argue on the Space Complexity of the Level-order traversal on a binary tree. → widest level — 5 CO1 PO1

c) — 4×3 CO3 PO2
   i. Argue on storing a binary tree using array-based vs pointer-based approach.
   ii. A perfect binary tree has 2^k nodes at depth k for $k = 0, 1, \ldots h$. Use this to prove that a perfect binary tree of height h has $2^{h+1} - 1$ nodes.
   iii. With a perfect binary tree of height h, if a node is randomly selected within the tree, what is the average length of the path from the root to that node?

3. a) Show that. 'If a node in a binary search tree has two children, then its successor has no left child and its predecessor has no right child'. Argue on the statement again considering the condition of having two children being withdrawn. — 10 CO3 PO2

b) Under what conditions would a pre-order and breadth-first traversal of a binary tree be the same? Is it possible to produce the same result for in-order and post-order traversal? → complete binary tree — 5 CO1 PO1

c) A set of number is stored in a Binary Search Tree. Design functions with least possible time complexity to satisfy the following requirements: — 10 CO2 PO2
   i. Print the numbers in descending order.
   ii. Given a number, return how many numbers are smaller than its value.


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